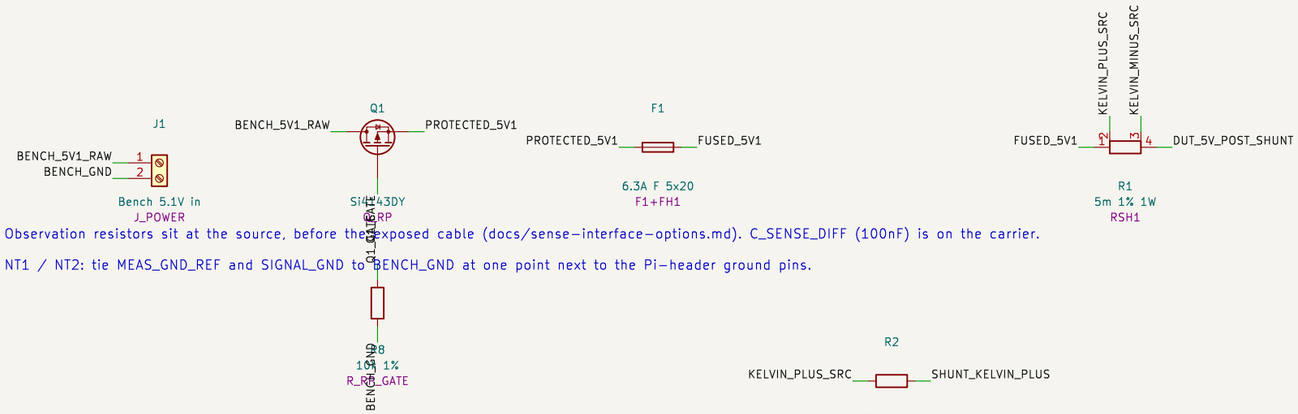
INTERPOSER — power path and observation taps (generated by gen_interposer_sch.py; do not edit by hand)

DUT current path: J1 -> Q1 (Si4143DY P-MOSFET reverse-polarity switch) -> F1 (cartridge in Schurter holder FH1) -> R1 shunt -> J2 pins 2/4 -> Pi -> J2 GND pins -> J1 pin 2

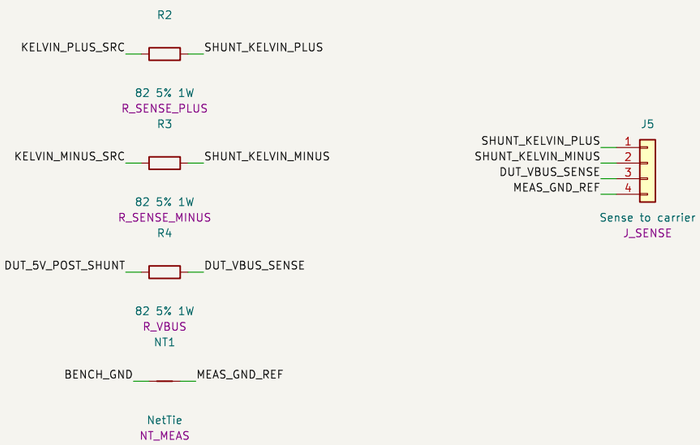
Everything upstream of R1 pin 1 is outside the DUT measurement. Q1 conducts only with correct bench polarity (gate held at ground by R8); it does not block reverse current from the Pi side.

R1 pins 2/3 are separate Kelvin pickups at the shunt lead pads; route KELVIN_* as a coupled pair, never through the 5A copper.

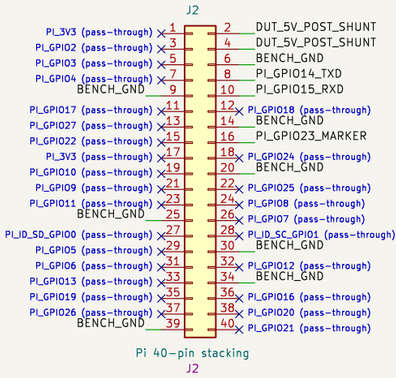


Observation resistors sit at the source, before the exposed cable (docs/sense-interface-options.md). C_SENSE_DIFF (100nF) is on the carrier.

NT1 / NT2: tie MEAS_GND_REF and SIGNAL_GND to BENCH_GND at one point next to the Pi-header ground pins.



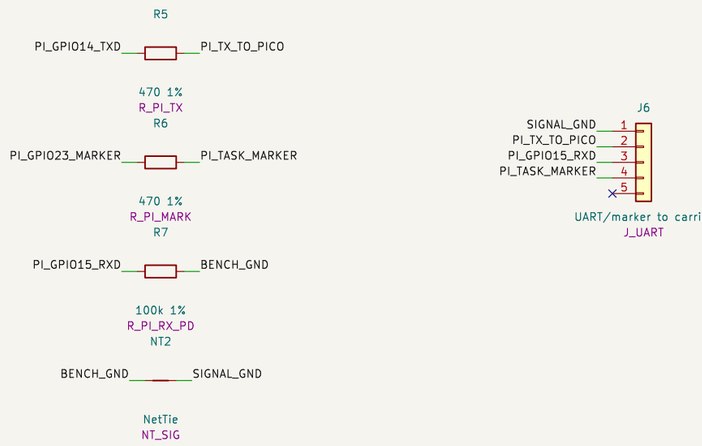
J2: all 40 contacts pass through unchanged to the HAT above



UART/marker (docs/uart-interface-qualification.md): 470R source resistors for Pi outputs here; Pico->Pi 470R and Pi-output 100k pull-downs are on the carrier.

R7 (100k Pi-RX pull-down) is on the interposer. Pi GPIO choice (TX GPIO14, RX GPIO15, marker GPIO23) is a default pending AI HAT confirmation.

The carrier pull-downs on PI_TX_TO_PICO/PI_TASK_MARKER and R7 load Pi pins under the approved UART-bias measurement-boundary exception. J6 pin 5 empty.



Prototype — not released

picopower-proto

Sheet: /

File: interposer.kicad_sch

Title: Interposer

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 1/1